AUTONOMOUS CYBER DEFENDER

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1 PROJECT SUMMARY

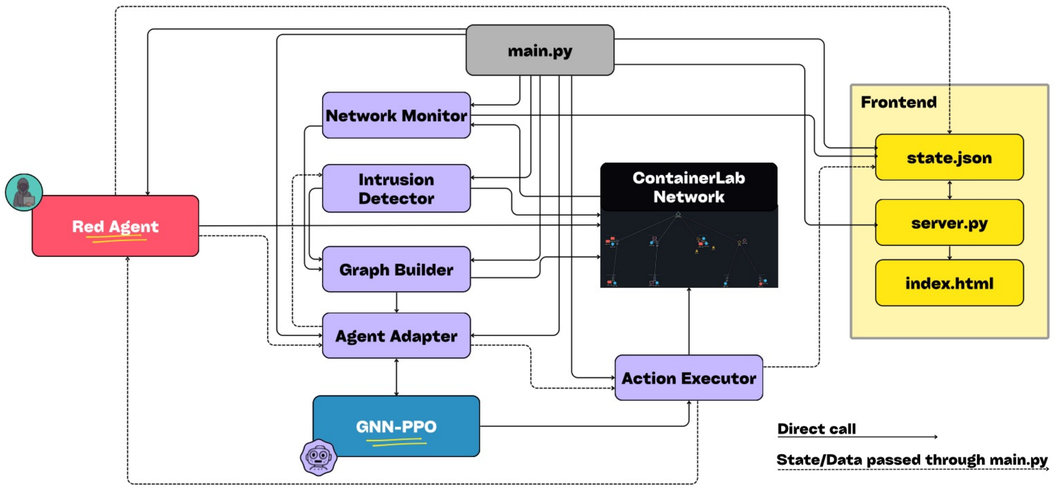
Cyber attacks are growing faster than human response. Simulation-trained agents help, but the sim-to-real gap complicates direct deployment on live infrastructure.

A graph-based PPO agent was built in **CybORG CAGE-4 (Simulation)** to defend a network, then deployed to a live **Containerlab (Real)** environment. A **red agent** attacks while a **blue agent** observes, reasons, and defends in real time, all on a real network.

2 HOW IT WORKS

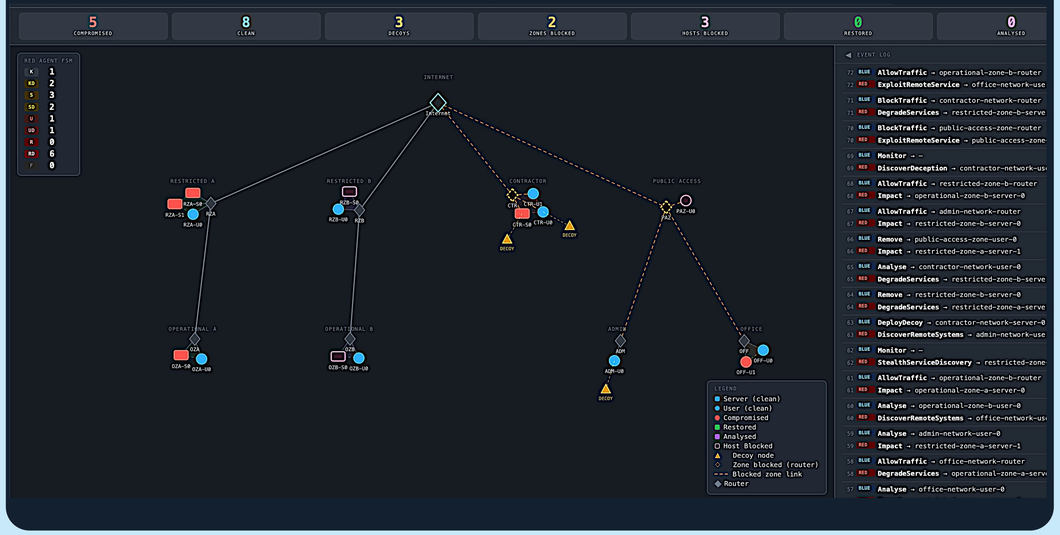
- A graph-based PPO agent was trained in CybORG CAGE-4 to detect and respond to threats in simulations.
- The agent was evaluated in simulation then connected to a live Containerlab network for real world testing.
- Live observations were passed to the agent in real time via a bridge pipeline.
- Decisions and performance of the bridge were compared against the simulation results.

SYSTEM ARCHITECTURE



LIVE DASHBOARD

Real-time node status and event log



TECHNOLOGIES

Agent & ML

- CybORG Cage-4
- PyTorch Geometric
- RLlib PPO
- PyTorch

Infrastructure & Tools

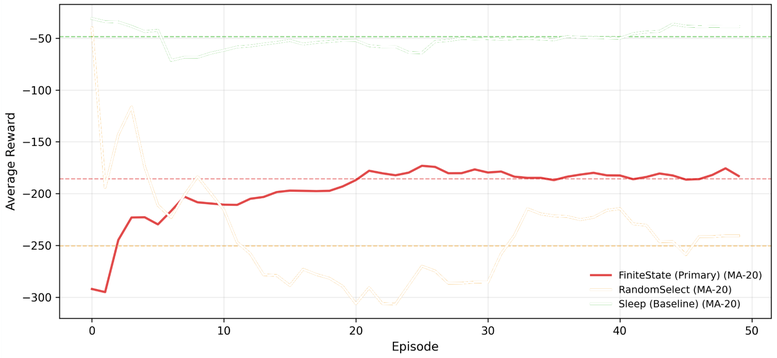
- Orbstack
- Docker
- Containerlab
- Alpine Linux
- Python

Testing & CI

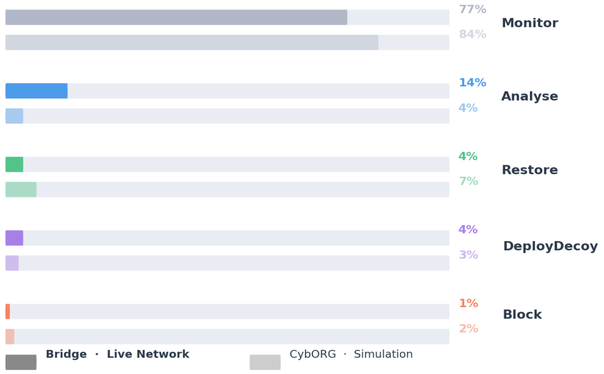
- pytest
- GitHub Actions
- coverage.py

RESULTS & EVALUATION

Smoothed reward over 50 episodes:
Blue Agent performance vs **Red Agent** in simulation



Blue Agent action distribution:
Bridge (live network) vs CybORG (simulation)



3 CONCLUSION

Project shows that some learned behaviours transfer successfully without retraining, demonstrating the potential of reinforcement learning for cyber defence, but with some limitations.

Transfer is Achievable

Action rates matched closely in both environments, confirming the GNN's policy generalises beyond its training environment.

Observation Sparsity

Analyse actions 3.5× higher in bridge. Policy probes more before committing when observations are sparse.

Conservative Policy

Restore 7× lower in bridge. 81% Monitor strategy cannot keep pace with a fast-escalating structured attacker in real deployment.